Clock generator & Reset fanout

File: clock_gen.kicad_sch

Direct 4-lane PCIe

File: direct_PCIe.kicad_sch

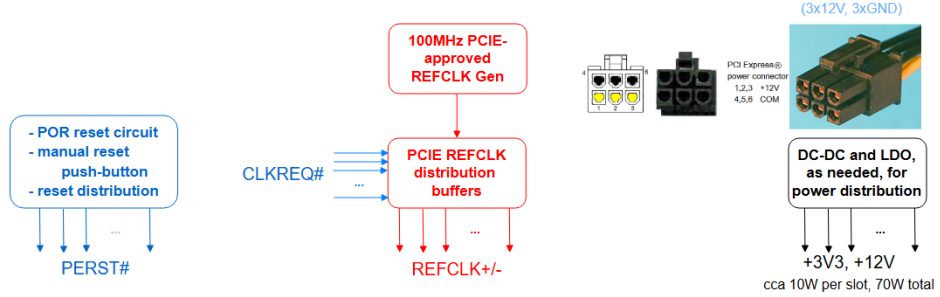
PCIe switch

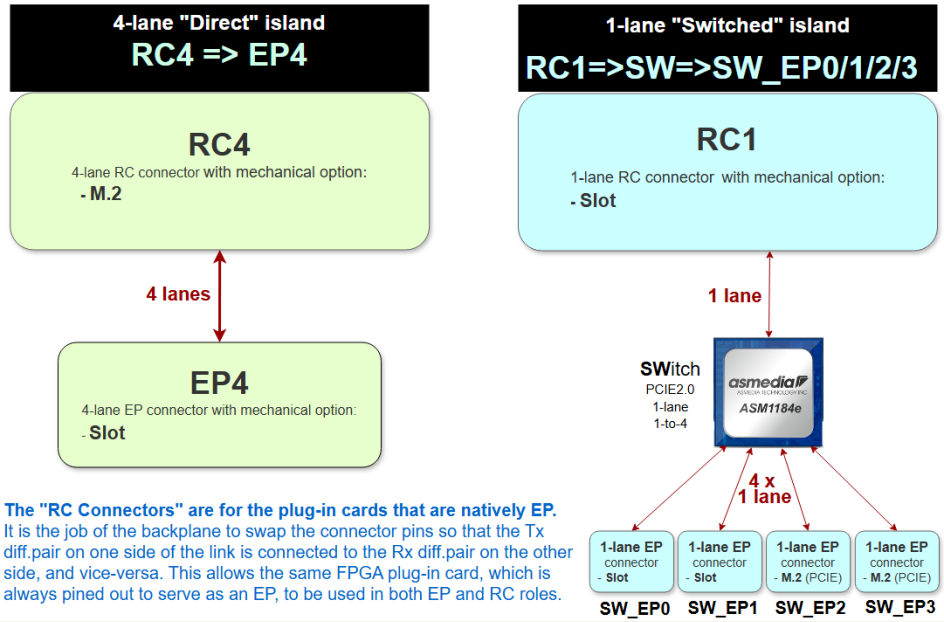
File: pcie_switch.kicad_sch

Switched 4 x 1-lanes PCIe

File: 4_1-lanes_PCIe.kicad_sch

PCIE mini Backplane





The "RC Connectors" are for the plug-in cards that are natively EP.
It is the job of the backplane to swap the connector pins so that the Tx diff.pair on one side of the link is connected to the Rx diff.pair on the other side, and vice-versa. This allows the same FPGA plug-in card, which is always pined out to serve as an EP, to be used in both EP and RC roles.

APPROVALS MAY BE VIEWED IN PLM SYSTEM					
REV	ECO	DESCRIPTION	DRAWN BY	APPROVED BY	DATE

ELECTRONIC APPROVALS:

ALL APPROVALS FOR THIS DRAWING, ASSOCIATED PARTS LIST(S), AND REVISION CHANGES, ARE CONTROLLED IN THE PRODUCT

nlnet FOUNDATION

Envox Experimental Zone

PCIe2 RootC NGI ZERO

KiCad

open EMS

open source hardware

OSHW – Open Source HardWare: https://oshwa.org
Licensed under CERN-OHL-P v2
https://cern-ohl.web.cern.ch
COPYRIGHT (C) CHILI.CHIPS.LLC

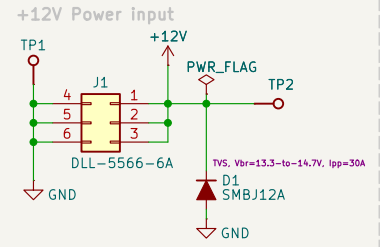
Sheet: /
File: openpci2-backplane.kicad_sch

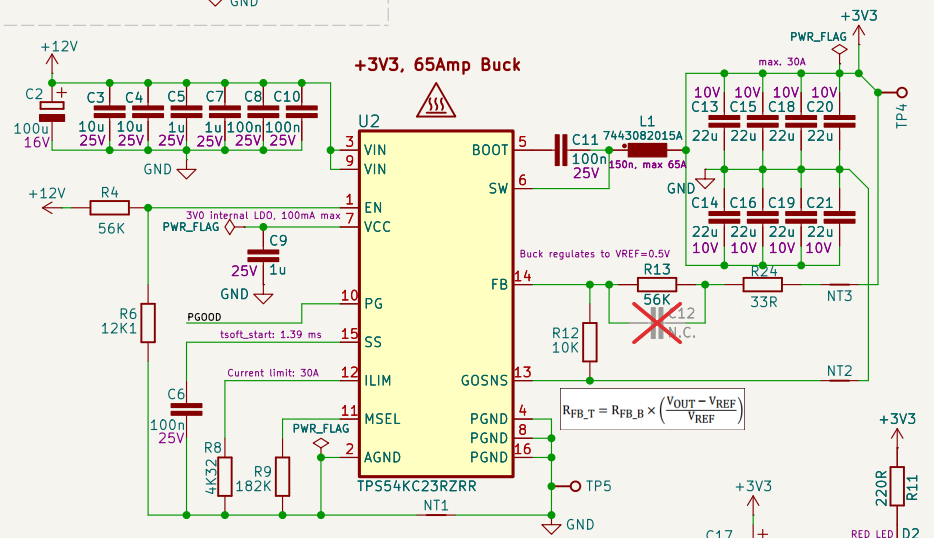
Title: openPCIE2 Backplane

Size: A4	Date: 2026-10-06	Rev: RevB
KiCad E.D.A. 10.0.7		Id: 1/5

H1 H2 H3 H4 H5 H6 H7 H8 H9 H10 H11 H12

Power Supply

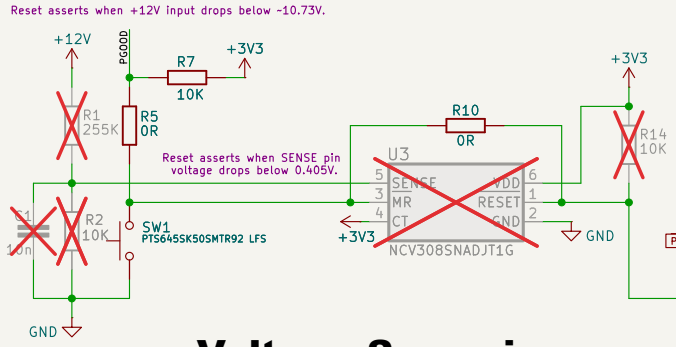


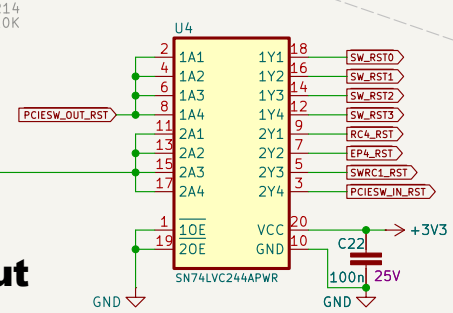


1*Ramp	Relative Ramp Amplitude VREF = 0.5V	Zero Location (kHz)
RAMP1	1×	32
RAMP2	1.8×	32
RAMP3	1.6×	53
RAMP4	3.0×	53

14	FB	I	Output voltage feedback input. A resistor divider from the output voltage to GOSNS (tapped to FB pin) sets the output voltage. Connect the FB divider to the output voltage near the load.
13	GOSNS	I	Negative input of the differential remote sense circuit. Connect to a ground sense point near the load.

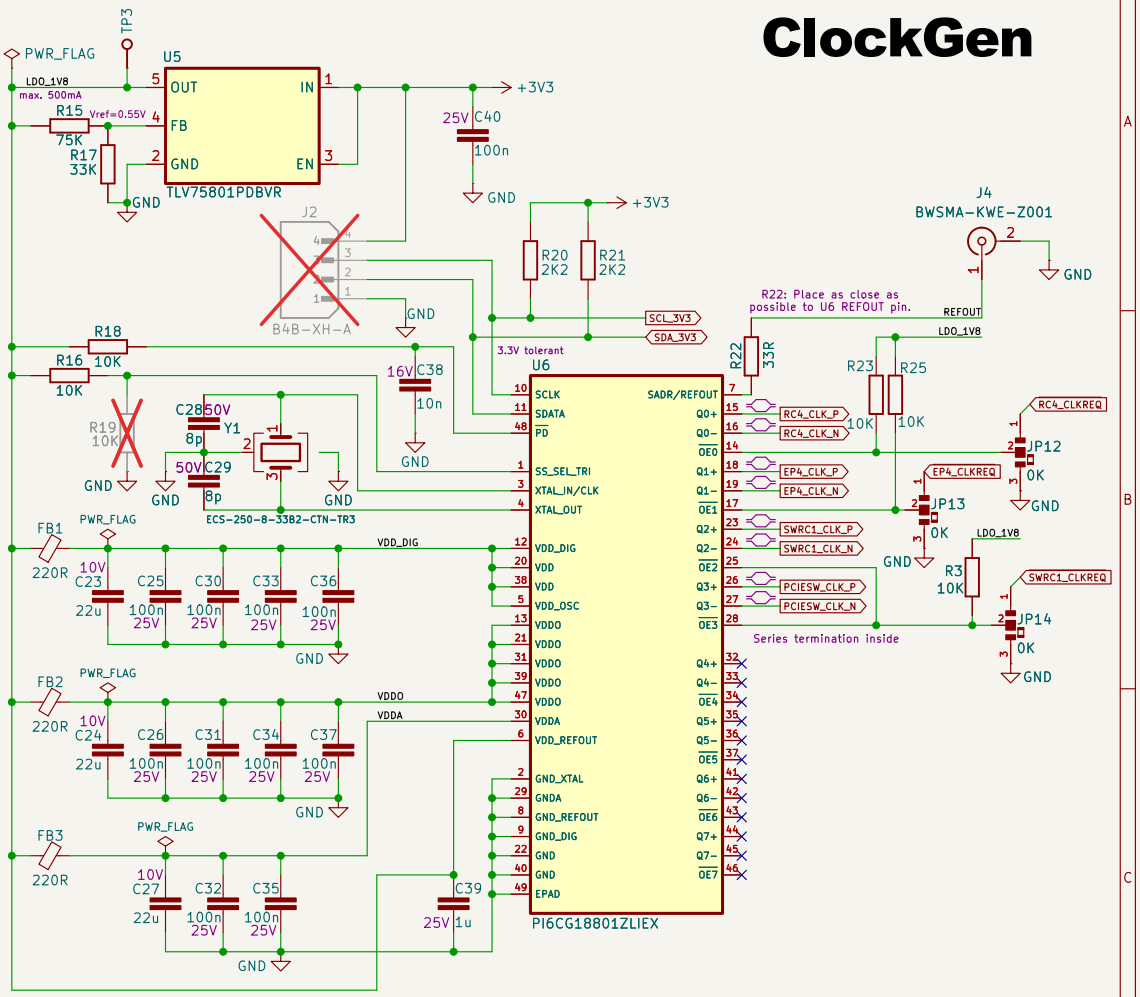
MSEL PIN RESISTANCE TO AGND (kΩ)	OPERATION MODE UNDER LIGHT LOAD	SWITCHING FREQUENCY (f_{SW}) (kHz)	RAMP
182	Skip-mode	1400	RAMP4





Voltage Supervisor
Manual Reset, Reset Fanout

ClockGen



OSHW – Open Source HardWare: https://oshwa.org
Licensed under CERN-OHL-P v2
https://cern-ohl.web.cern.ch
COPYRIGHT (C) CHILI.CHIPS.LLC

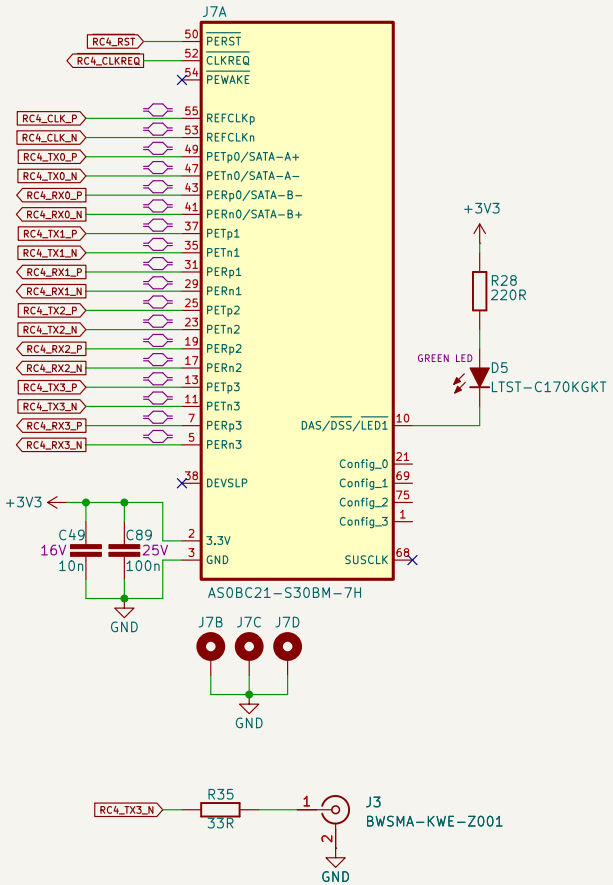
Sheet: /Clock generator & Reset fanout/
File: clock_gen.kicad_sch

Title: openPCIE2 Backplane

Size: A4	Date: 2026-10-06	**Rev: RevB**
KiCad E.D.A. 10.0.7		Id: 2/5

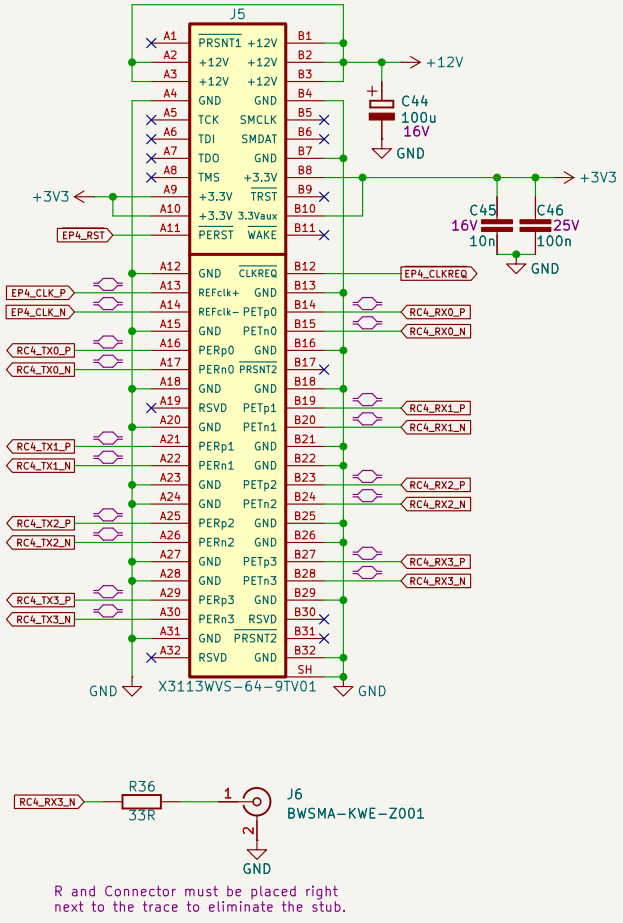
Direct Connection Island

RC4



R and Connector must be placed right next to the trace to eliminate the stub.

EP4



R and Connector must be placed right next to the trace to eliminate the stub.

PCIEx1 pinout

Pin	Side B	Side A	Description
1	+12 V	~~PRSNT1#~~	Must connect to farthest PRSNT2# pin
2	+12 V	+12 V	Main power pins
3	+12 V	+12 V	
4	Ground	Ground	
5	~~SMCLK~~	~~TCK~~	SMBus and JTAG port pins
6	~~SMDAT~~	~~TDI~~	
7	Ground	~~TDO~~	
8	+3.3 V	~~TMS~~	
9	~~TRST#~~	+3.3 V	
10	+3.3 V aux	+3.3 V	Aux power & Standby power
11	~~WAKE#~~	PERST#	Link reactivation; fundamental reset [25]
Key notch			
12	CLKREQ# [26]	Ground	Clock Request Signal
13	Ground	REFCLK+	Reference clock differential pair
14	HSOp(0)	REFCLK-	
15	HSOn(0)	Ground	Lane 0 transmit data, + and −
16	Ground	HSIp(0)	
17	~~PRSNT2#~~	HSIn(0)	Lane 0 receive data, + and −
18	Ground	Ground	

PCI Express x1 cards end at pin 18

Legend	
Ground pin	Zero volt reference
Power pin	Supplies power to the PCIe card
Card-to-host pin	Signal from the card to the motherboard
Host-to-card pin	Signal from the motherboard to the card
Open drain	May be pulled low or sensed by multiple cards
Sense pin	Tied together on card
Reserved	Not presently used, do not connect

OSHW – Open Source HardWare: https://oshwa.org
Licensed under CERN-OHL-P v2
https://cern-ohl.web.cern.ch
COPYRIGHT (C) CHILI.CHIPS.LLC

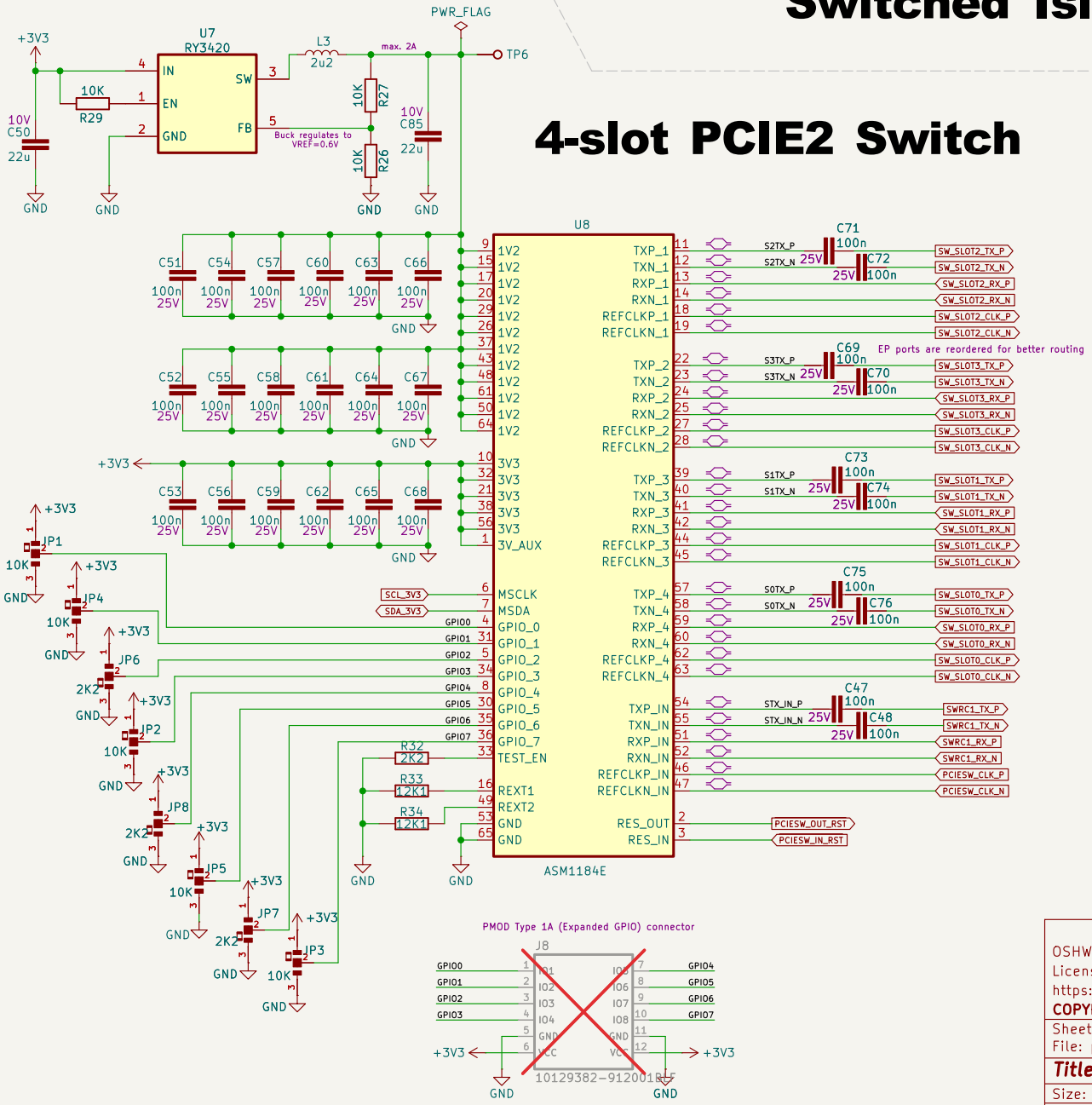
Sheet: /Direct 4-lane PCIe/
File: direct_PCIe.kicad_sch

Title: openPCIE2 Backplane

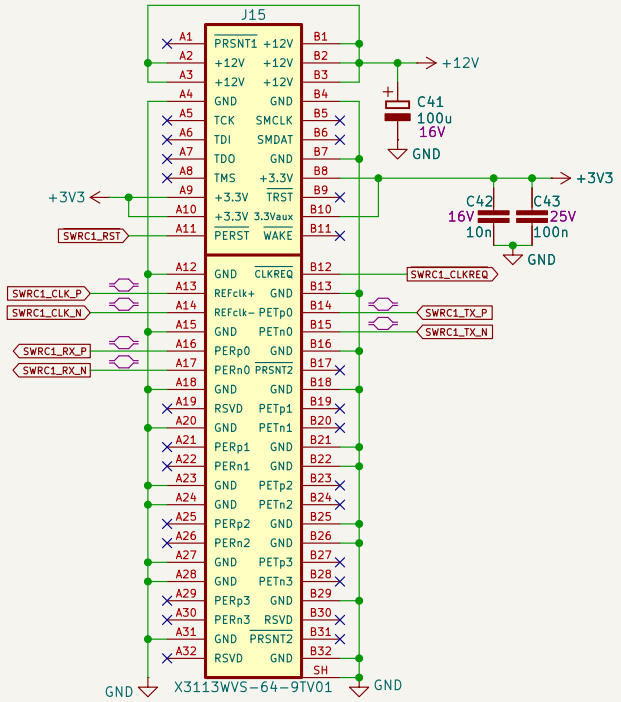
Size: A4	Date: 2026-10-06	**Rev: RevB**
KiCad E.D.A. 10.0.7		Id: 3/5

Switched Island (Part 1)

4-slot PCIE2 Switch



SWRC1



OSHW – Open Source HardWare: https://oshwa.org
Licensed under CERN-OHL-P v2
https://cern-ohl.web.cern.ch
COPYRIGHT (C) CHILI.CHIPS.LLC

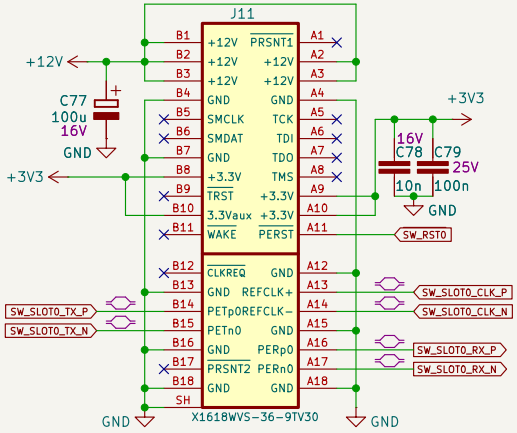
Sheet: /PCIe switch/
File: pcie_switch.kicad_sch

Title: openPCIE2 Backplane

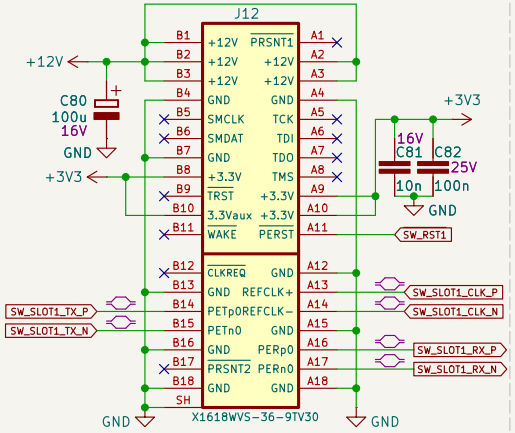
Size: A4	Date: 2026-10-06	**Rev: RevB**
KiCad E.D.A. 10.0.7		Id: 4/5

Switched Island, Part 2

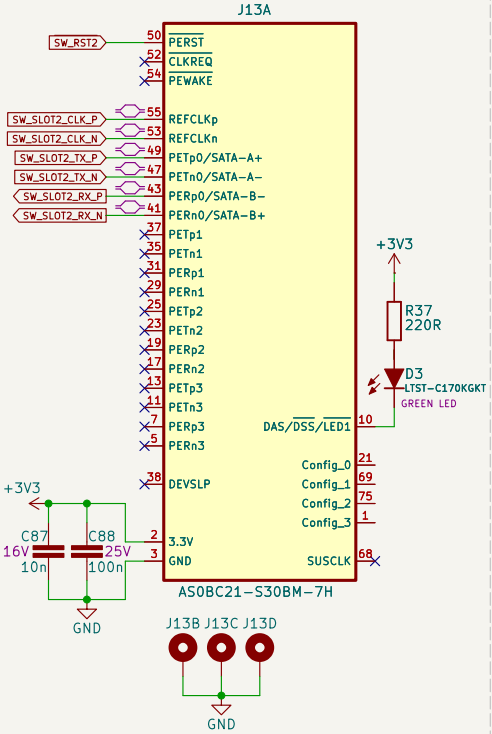
SW_EP0



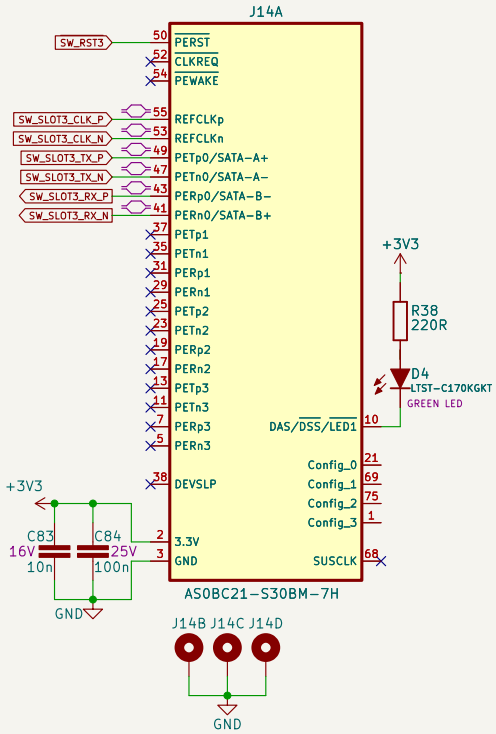
SW_EP1



SW_EP2



SW_EP3



OSHW – Open Source HardWare: https://oshwa.org
Licensed under CERN-OHL-P v2
https://cern-ohl.web.cern.ch
COPYRIGHT (C) CHILI.CHIPS.LLC

Sheet: /Switched 4 x 1-lanes PCIe/
File: 4_1-lanes_PCIe.kicad_sch

Title: openPCIE2 Backplane

Size: A4	Date: 2026-10-06	**Rev: RevB**
KiCad E.D.A. 10.0.7		Id: 5/5